



Phase structure and electrical properties of ternary PSN-PMN-PT ceramics and their applications in ultrasonic transducers

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ABSTRACT

In this work, a series of x PSN-(1 - x - y)PMN- y PT ($x = 0.06 - 0.43$, $y = 0.32 - 0.42$) ternary ceramics with different phase structures were synthesized by the solid state reaction method, and the specific phase structure, electrical properties and thermal stability as a function of compositions were investigated. It is found that with the increasing PT content, the phase transition sequence is $M_C + R \rightarrow M_C + M_A + R \rightarrow M_C + T$ for the ceramics with the same PSN content. Meanwhile, with the increasing PSN content, the MPB compositions shift to higher PT content. The maximum d_{33} value for $x = 0.06, 0.15, 0.29$ and 0.43 ceramics occurs at $y = 0.35, 0.37, 0.39$ and 0.40 , respectively, which are dominated by T phase and coexist with a small fraction of M_C phase and the corresponding d_{33} value is 665, 620, 585 and 650 pC/N. From 30 °C to the depolarization temperature T_d of each ceramic, the variation rate of d_{33} is 13 %, 7 %, 5 %, and 15 %, respectively, demonstrating that these ceramics have a good thermal stability. In addition, the 0.06PSN-0.59PMN-0.35PT ceramic with the highest d_{33} is selected as an active element of ultrasonic transducer, and the ultrasonic transducer with a center frequency of 4.65 MHz was produced. The -6 dB bandwidth and two-way insertion loss are 77.4 % and -21.45 dB, respectively. This work has important implications for the further development of high-performance PT based ferroelectrics.

1. Introduction

Piezoelectric materials have been widely used in industrial manufacturing, medical diagnosis and other fields, which can be attributed to the mutual conversion of mechanical energy and electrical energy [1–6]. Constructing materials with morphotropic phase boundary (MPB) compositions is a simple and practical method to effectively improve the piezoelectric response of ferroelectric ceramics, and it has been widely used in both lead-free [7,8] and lead-based systems [5,6,9]. Lead-based relaxation ferroelectric materials have been the subject of much research in recent decades because of their excellent piezoelectricity and dielectricity. Among them, the representative Pb(Mg_{1/3}Nb_{2/3})O₃-PbTiO₃ (PMN-PT) relaxor ferroelectrics with MPB compositions have been widely studied because of their extremely high electromechanical coupling coefficient ($k_{33} > 0.65$) and piezoelectric constant ($d_{33} > 600$ pC/N) [9–14]. However, the low phase transition

temperature T_{r-t} (60 – 90 °C) and Curie temperature T_c (130 – 160 °C) restrict their application in high-temperature environments [13]. For the development of new relaxor ferroelectrics with high T_c and T_{r-t} , the researchers introduced end elements with high T_c into PMN-PT, obtaining many new ternary systems with rising T_c ($T_c \geq 200$ °C) and T_{r-t} ($T_{r-t} \geq 100$ °C), such as Pb(Yb_{1/2}Nb_{1/2})O₃-Pb(Mg_{1/3}Nb_{2/3})O₃-PbTiO₃ (PYN-PMN-PT) [14], Pb(Mg_{1/3}Nb_{2/3})O₃-PbZrO₃-PbTiO₃ (PMN-PZ-PT) [15], Pb(In_{1/2}Nb_{1/2})O₃-Pb(Mg_{1/3}Nb_{2/3})O₃-PbTiO₃ (PIN-PMN-PT) [16], etc. Moreover, in comparison to PMN-PT systems, the T_c and d_{33} values of the ternary system can be regulated over a wider range, which is advantageous for users to select various components to satisfy the requirements of various applications. For instance, by adjusting the composition, the T_c of PYN-PMN-PT can vary between 165 and 370 °C, and the d_{33} can vary between 373 and 505 pC/N [14].

For PMN-PT based materials, the selection of end elements has a great influence on the performance. Among all relaxor ferroelectric, Pb

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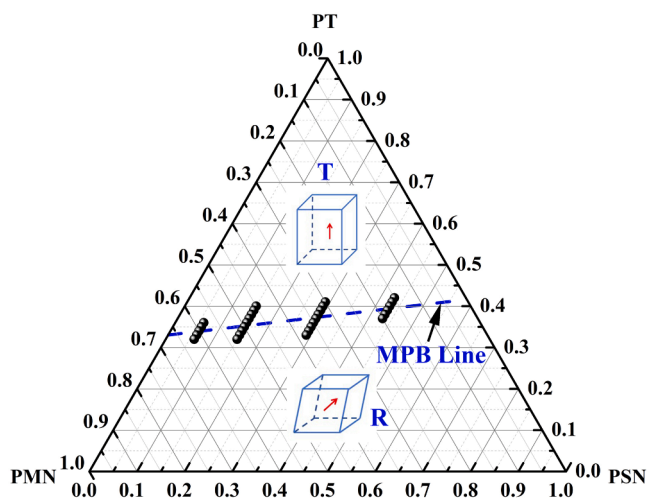


Fig. 1. A phase diagram of PSN-PMN-PT ternary system. The black points are the compositions in this paper.

($\text{Sc}_{1/2}\text{Nb}_{1/2}\text{O}_3$ (PSN) is a typical material with a rhombohedral phase structure and its T_c is about 90 °C [17]. The Rhombohedral-Tetragonal MPB of PSN-PT binary system had been successfully constructed near 42 mol% PT, at which high T_c , piezoelectric and dielectric properties

have been obtained [18]. By combining the MPBs of PMN-PT and PSN-PT, a PSN-PMN-PT ternary system was developed by Yamashita et al. and the d_{33} and T_c reach 640 pC/N and 205 °C, respectively, in 0.29PSN-0.34PMN-0.37PT composition [19], which are superior to the PMN-PT system. Recently, the improved thermal stability and piezoelectric performances of PSN-PMN-PT through composition design, ion doping and composite method [20–23]. As reported by Guo et al., the d_{33} and T_c of 0.43PSN-0.17PMN-0.40PT composition are 650 pC/N and 255 °C, respectively, in the meantime, the variation rate of d_{33} remains below 20 % within a temperature range of 30 – 250 °C [20]. Dong et al. conducted an investigation on 0.15PSN-(0.85 – x)PMN- x PT ceramics doped with 2 mol% Sm^{3+} , finding that the d_{33} reaches 815 pC/N when $x = 0.33$, and the d_{33} varies by only 8 % over the test temperature range of 30 – 110 °C [21]. Guo et al. noted that the composite ceramic of 0.15PSN-0.51PMN-0.34PT/0.05 mol ZnO exhibits an improved d_{33} (590 pC/N) compared to pure 0.15PSN-0.51PMN-0.34PT ($d_{33} = 529$ pC/N), and after annealing at 200 °C, the retained d_{33} maintains 92 % of the value before annealing [22]. The results above mentioned show that PSN-PMN-PT system has great potential for practical applications. Nevertheless, we notice that there is a lack of systematic studies on how the phase structure evolves with composition, and the impact of phase structure evolution on electrical performance. These questions are essential to comprehend the physical principles underlying the excellent performance of this system, which are very critical for the further development of PSN-PMN-PT solid solution as well as the component

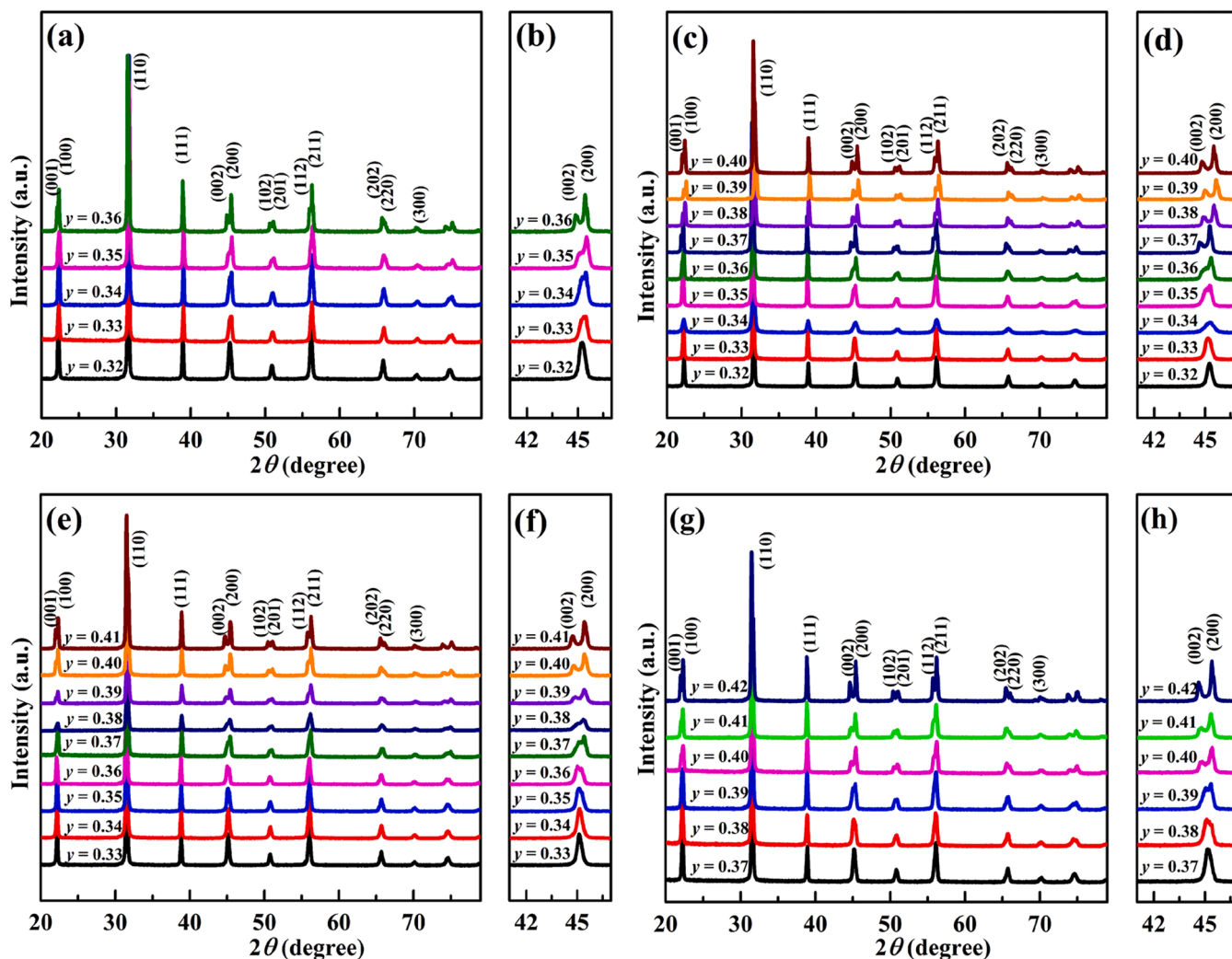


Fig. 2. XRD patterns for $x\text{PSN}-(1 - x - y)\text{PMN}-y\text{PT}$ ceramics: (a)-(b) $x = 0.06$, (c)-(d) $x = 0.15$, (e)-(f) $x = 0.29$, (g)-(h) $x = 0.43$.

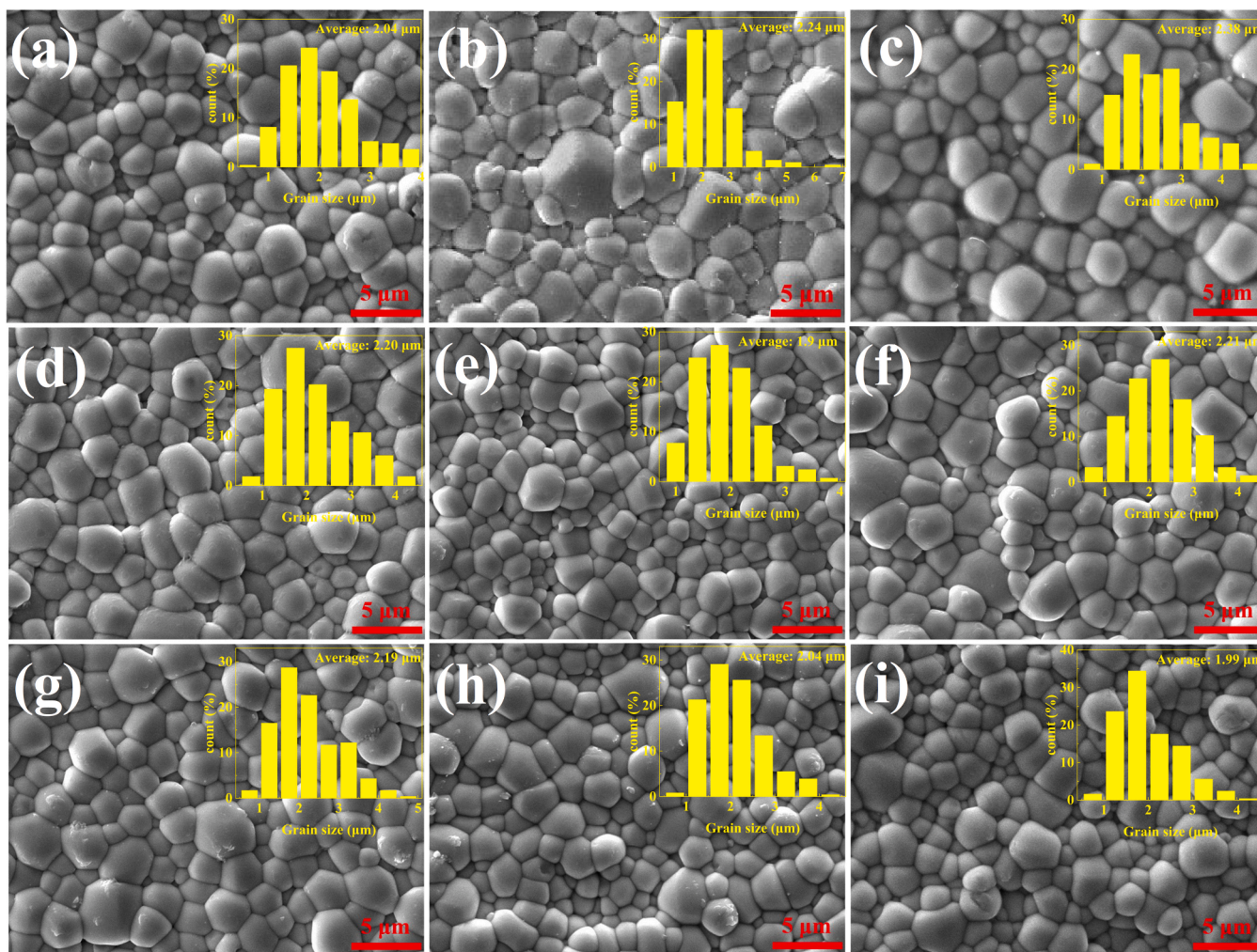


Fig. 3. Surface micromorphology of 0.15PSN-(0.85 - y)PMN- y PT ceramics: (a)-(i) $y = 0.32 - 0.40$.

design for other ternary systems.

In this study, the ternary x PSN-(1 - x - y)PMN- y PT ($x = 0.06 - 0.43$, $y = 0.32 - 0.42$) ceramics with different phase structures (rhombohedral, tetragonal and monoclinic phases) were established via the linear combination rule, as shown in Fig. 1. The phase structure, electrical properties and thermal stability of PSN-PMN-PT ceramics were analyzed in detail. Furthermore, to evaluate the potential in applications, an ultrasonic transducer with a center frequency (f_c) of 4.65 MHz was fabricated using PSN-PMN-PT ceramic as an active element and the -6 dB bandwidth (BW) and two-way insertion loss (IL) were measured.

2. Experimental procedures

The PSN-PMN-PT ternary ceramics were prepared via the solid state reaction method. MgO (99.9 %), Sc_2O_3 (99.99 %) and Nb_2O_5 (99.5 %) were used to synthesize the precursors $MgNb_2O_6$ (fabricated at 1000 °C) and $ScNbO_4$ (fabricated at 1150 °C). The $MgNb_2O_6$, $ScNbO_4$, TiO_2 (98 %) and PbO (99.99 %) powders were selected and weighed in accordance with the stoichiometric proportion and 2 mol% PbO extra was added to make up for the vaporization of PbO during sintering. After that, alcohol and the above oxide powders were mixed in a 1:1 ratio by wet-milling for 24 h. Next, the combined powder was subjected to calcination at 800 - 850 °C for 2 h, the calcined mixture was mixed again for 12 h. Subsequently, the powder was conglutinated with a solution of polyvinyl alcohol (PVA) and pressed into tablets under 8 MPa, and then the PVA was burned off at 550 °C for a duration of 2 h. Finally,

the ceramic tablets were covered by powders with the same composition to maintain a Pb-rich atmosphere and sintered at 1200 - 1280 °C for 4 h. The polished samples (cylindrical disks with diameters of 11.40 - 11.54 mm and thickness of 0.74 - 0.90 mm) were coated with silver electrodes on both sides in preparation for electrical tests. The poling temperature, duration and electric field are 130 °C, 15 min and 30 kV/cm, respectively.

The phase structure and morphology of the samples were examined by X-ray diffraction (Bruker D2 PHASER Gen2, Germany) and scanning electron microscope (TESCAN, VEGA II XMU, Czechia), respectively. The Archimedes method was utilized to measure the bulk density of the sintered samples. The d_{33} was tested by a quasi-static piezoelectric meter (ZJ-6A, China). For poled samples, a LCR meter (Agilent E4980A, USA) was used to test the dielectric permittivity (ϵ_r) and loss ($\tan\delta$) as a function of temperature from 30 to 300 °C. The ferroelectric polarization hysteresis behaviors ($P - E$) were obtained by a ferroelectric test system (aixACCTF Analyzer 2000, Germany). With the use of impedance analyzer (Agilent 4294 A, USA), the electromechanical coupling factor k_p was tested. The thermal stimulated depolarization current was measured by standard short circuit approach through Keithley 6487. The temperature stability of d_{33} was measured using the length-extensional mode of k_{33} piezoelectric resonator via resonant method, the detailed experimental process is shown in Note S1.

The 0.06PSN-0.59PMN-0.35PT ceramic with the highest piezoelectric performance was used as piezoelectric elements to prepare ultrasonic transducer. Fig. 10(a) displays the structure diagram of the

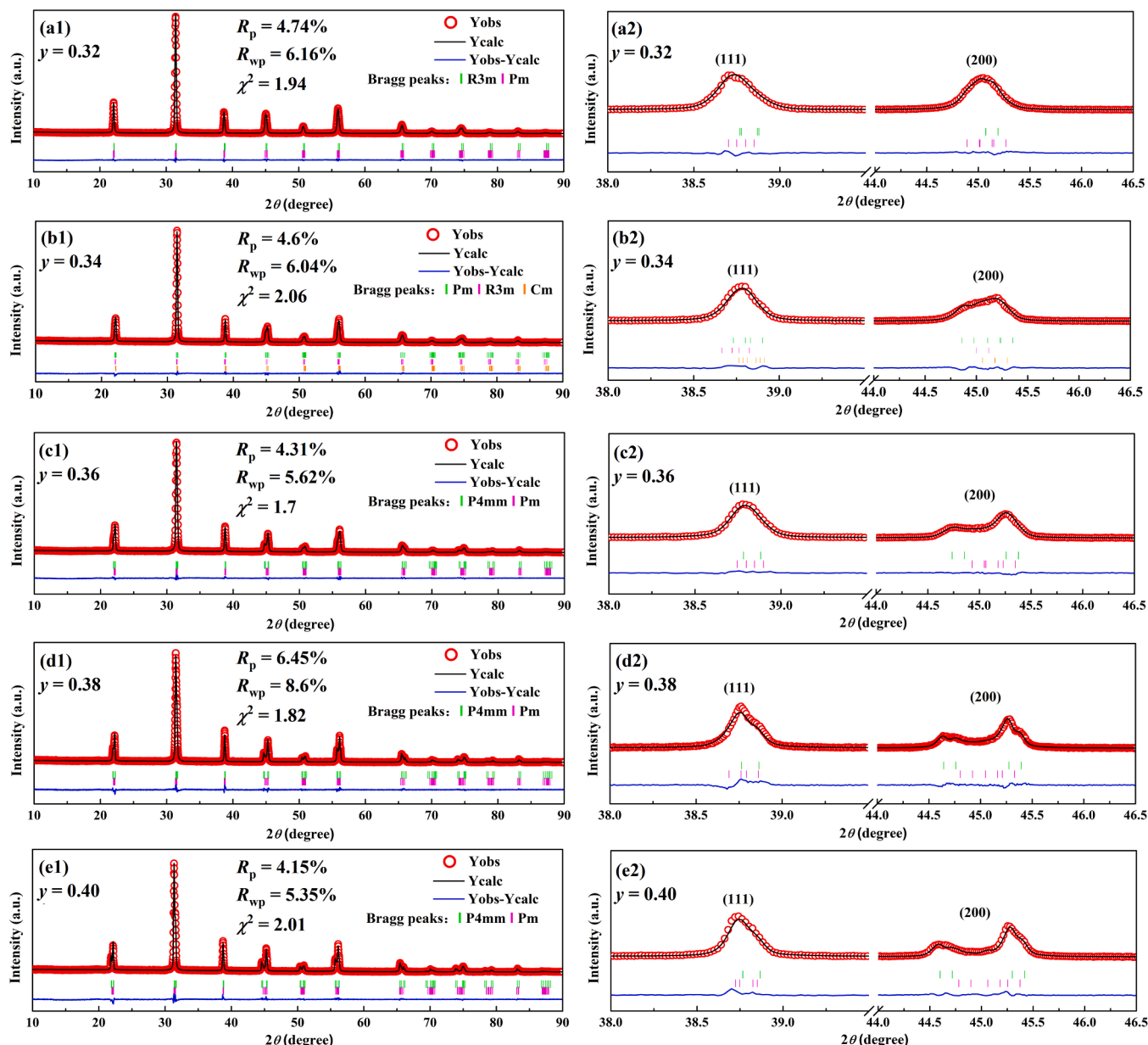


Fig. 4. Rietveld refinement for the 0.15PSN-(0.85 - y)PMN-yPT at room temperature: (a)-(e) $y = 0.32, 0.34, 0.36, 0.38, 0.40$.

ultrasonic transducer, which is made up of active elements, matching layers, backing, wires, and casing. The matching layer adopts a double-layer matching layer structure, the matching layers 1 and 2 are made of AZ31B Mg alloy and Epo-Tek 301, respectively. The main fabrication procedures are listed in Note S2. The performance of the PSN-PMN-PT transducer was assessed by the traditional pulse-echo response measurement technique [24,25]. The detailed equation of center frequency (f_c), BW and IL are shown in Note S3.

3. Results and discussion

Fig. 2 displays the XRD patterns of x PSN-(1 - x - y)PMN-yPT ceramics. Each sample is of pure perovskite structure without pyrochlore phase, which implies that PSN and PMN-PT have successfully formed a solid solution. For PT-based ferroelectric systems, the change of the phase structure can be generally reflected through the diffraction peak of (200) around 45° [26,27]. Fig. 2(b), (d), (f) and (h) depict the evolution of (200) diffraction peak with the variable composition. It is evident that with the increasing PT content, the same variation trend of

the (200) peak is observed for PSN-PMN-PT samples with different x ($x = 0.06, 0.15, 0.29$, and 0.43). Taking the $x = 0.15$ ceramics as an example, when the PT content is less than 0.35 mol ($y < 0.35$), the (200) diffraction peak is single and the crystalline structure of these compositions is dominated by rhombohedral phase. As y increases to 0.35 , the (200) peak starts to broaden, indicating that the phase transition begins in this component. When y further increases, a shoulder on the left side of the (200) peak occurs and its relative intensity gradually increases. As y rises to 0.40 , the (200) peak almost splits into (200) and (002), suggesting that the tetragonal phase is dominant.

For PT-based ferroelectric ceramics, grain size is one of the key factors affecting their electric properties, because the grain size determines the domain size, which is closely related to electric properties [28–30]. As an example, Fig. 3 presents the surface SEM images of the 0.15PSN-(0.85 - y)PMN-yPT, and illustrate the corresponding grain size distribution. It is evident that the all as-prepared samples exhibit a good dense structure with no evident pores. The bulk density of these ceramics varies between 7.5 and 7.85 g/cm³, and the relative densities are greater than 95 %. According to the grain size distribution in Fig. 3, it

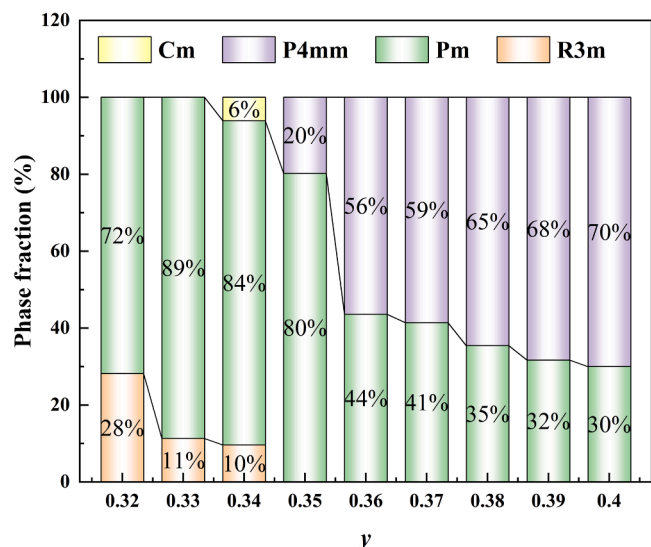


Fig. 5. The phase fraction of 0.15PSN-(0.85 - y)PMN-yPT with PT content.

has been observed that the mean grain size for all 0.15PSN-(0.85 - y)PMN-yPT ceramics varies within a small range of 1.9 - 2.38 μm , indicating the influence of grain size on the electric performances of $x = 0.15$ ceramics with different PT content can be ignored. This conclusion also applies to $x = 0.06$, 0.29, and 0.43 ceramics with different PT content in this study.

To further determine the composition-induced phase transition behaviors of the PSN-PMN-PT ceramics, the fitting of the XRD patterns of $x = 0.15$ ceramics were fitted using the Rietveld refinement [31], the refined results are presented in Fig. 4 and Fig. S1. As anticipated, the peak positions and intensities of the fitted profiles match well with the measured results for all ceramics, indicating that the refinement results are realistic and credible. The optimal Rietveld refinement results are achieved for all $x = 0.15$ ceramics by using the following phase models: R3m + Pm (R + M_C) for $y = 0.32 - 0.33$, R3m + Pm + Cm (R + M_C + M_A) for $y = 0.34$, and P4mm + Pm (T + M_C) for $y = 0.35 - 0.40$, demonstrating that multi-phases with competing symmetries coexist in the PSN-PMN-PT system. The detailed phase volume fractions of all $x = 0.15$ ceramics based on the refined results is exhibited in Fig. 5. Clearly, in the ceramics with $y = 0.32$ and 0.33, the R and M_C phases coexist and M_C phase is dominant, and the M_C phase fraction is 72 % and 89 %, respectively. For $y = 0.34$ ceramic, the three-phase coexistence of M_C , M_A and R is observed, with volume fractions of 84 %, 6 % and 10 %, respectively. When the y exceeds 0.34, R and M_A phases disappear, and the samples show the two-phase coexistence of M_C and T. Moreover, as the y rises from 0.35 to 0.40, the T phase fraction progressively increased from 20 % to 70 % and the M_C phase fraction decreased from 80 % to 30 %. The results of the phase fraction for $x = 0.15$ ceramics are consistent with the analysis in Fig. 2(b) and are also in agreement with the phase diagram (Fig. 1). Previous studies [18] have confirmed that the 0.43PSN-0.17PMN-0.40PT ceramic consists of a combination of M_C (21.19 %) and T (78.81 %) phases and T phase is dominant, implying that similar trend of phase fraction with increased PT content should be obtained in $x = 0.06$, 0.29 and 0.43 components. The multi-phases coexistence is an intrinsic feature in PSN-PMN-PT system, which is in good agreement with the observations in other PT-based ferroelectrics [32-34].

The variations of the dielectric permittivity ϵ_r and dielectric loss factor $\tan\delta$ for poled xPSN-(1 - x - y)PMN-yPT ceramics with temperature in the range of 30 - 300 $^{\circ}\text{C}$ are illustrated in Fig. 6(a)-(d). All measurements of the curves are measured at a frequency of 1 kHz. As can be observed, for $x = 0.06$, 0.15, 0.29 and 0.43 ceramics with low PT content, the presence of two distinct dielectric anomalous peaks is

observed, corresponding to the phase transition of ferroelectric-ferroelectric (F-F) and ferroelectric-paraelectric (F-P), respectively. With the increasing PT content, the dielectric anomalous peak associated with F-F phase transition (at low temperature T_{F-F}) gradually weakens and only a single dielectric anomalous peak connected with F-P phase transition (at high temperature T_C) is obviously observed. The visual phase transition temperatures T_{F-F} and T_C for all xPSN-(1 - x - y)PMN-yPT samples are depicted in Fig. 6(e). In all ceramics with the same x , the T_{F-F} decreases while T_C increases with increasing PT content, which can be ascribed to gradually increased tetragonal phase fraction, since the PT is of tetragonal phase as well as a high T_C of 490 $^{\circ}\text{C}$ [34]. Besides, the observation indicates that, for the ceramics with the same PT content, the PSN-PMN-PT system exhibits an increase in both T_{F-F} and T_C as the PSN content increases, which results from the larger rhombohedral lattice deformation and higher T_C of PSN in comparison with PMN [35]. The detailed T_{F-F} and T_C of all the ceramics are given in Table 1. It should be noted that the $\tan\delta$ of all ceramics is less than 0.05 in the whole test temperature range.

Additionally, as the PT content increases, the dielectric anomaly peak connected with F-P phase transition in $x = 0.06$, 0.15, 0.29 and 0.43 ceramics gradually become narrow, indicating the relaxation behavior gradually weakens. Such dielectric relaxor behavior can be demonstrated by the modified Curie-Weiss law [36] of $1/\epsilon - 1/\epsilon_{\text{max}} = (T - T_{\text{max}})^{\gamma}/C$, where C represents the Curie-Weiss constant, and γ represents the diffusion coefficient, and ϵ_{max} and T_{max} refer to the maximum dielectric constant and the corresponding temperature, respectively. The fitting results of the representative 0.15PSN-(0.85 - y)PMN-yPT ceramics are as depicted in Fig. 6(f). The calculated γ value decreases from 1.79 for $y = 0.32$ to 1.58 for $y = 0.40$ ceramics, which can be put down to the reason that PT is a classical ferroelectric. In the meantime, the high γ signifies the strong relaxor characteristics of the PSN-PMN-PT system, which can be attributed to the intensified ionic disorder and local charge mismatch at B site [37,38].

Fig. 7(a) shows the variation trend of relative permittivity ϵ_r of all components. As the PT content increases, the trend of ϵ_r can be observed to exhibit an initial increase followed by a subsequent decrease in each special x . The maximum value is obtained near $y = 0.35$, 0.38, 0.39 and 0.41, respectively, for $x = 0.06$, 0.15, 0.29 and 0.43 ceramics. In combination with XRD analysis, these samples have the two-phase coexistence of T + M_C and the T phase is dominant. The polarization-electric field ($P - E$) loops of xPSN-(1 - x - y)PMN-yPT samples under a 20 kV/cm electric field are depicted in Fig. S2. The presence of well-shaped hysteresis loops with saturation in all samples clearly indicates their well ferroelectricity. Fig. 7(b) displays the remanent polarization P_r and coercive electric field E_c of the xPSN-(1 - x - y)PMN-yPT ceramics. As y increases for each x , it is evident that the values of P_r slightly fluctuate around 30 $\mu\text{C}/\text{cm}^2$, while the E_c values gradually increase due to the increase of T phase fraction [39]. The detailed P_r and E_c of all the ceramics are listed in Table 1.

For piezoelectric materials, piezoelectric coefficient (d_{33}) is commonly regarded as a crucial parameter that signifies the capacity to transform external force into charge. In general, the d_{33} exhibits a direct correlation with the ferroelectric polarization and dielectric characteristics of piezoelectric ceramics [40]. The equation of $d_{33} \propto \epsilon_r P_r$ [33] can be used to establish the relationships among them. Fig. 7(c) and (d) present the $\epsilon_r P_r$ and d_{33} of xPSN-(1 - x - y)PMN-yPT ceramics. It is evident that as PT content increases, both d_{33} and $\epsilon_r P_r$ exhibit a similar tendency of initial increase followed by decrease, suggesting the improvement of d_{33} is indeed resulted from the enhanced $\epsilon_r P_r$. The maximum d_{33} value for each x occurs at 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT samples and the corresponding value is 665, 620, 585, and 650 pC/N, respectively. It should be pointed out that the high d_{33} for these ceramics mainly originates from enhanced ϵ_r instead of P_r , since the P_r only presents a slight change. Combining the Rietveld refinement results of $x = 0.15$, it can be inferred that the high d_{33}

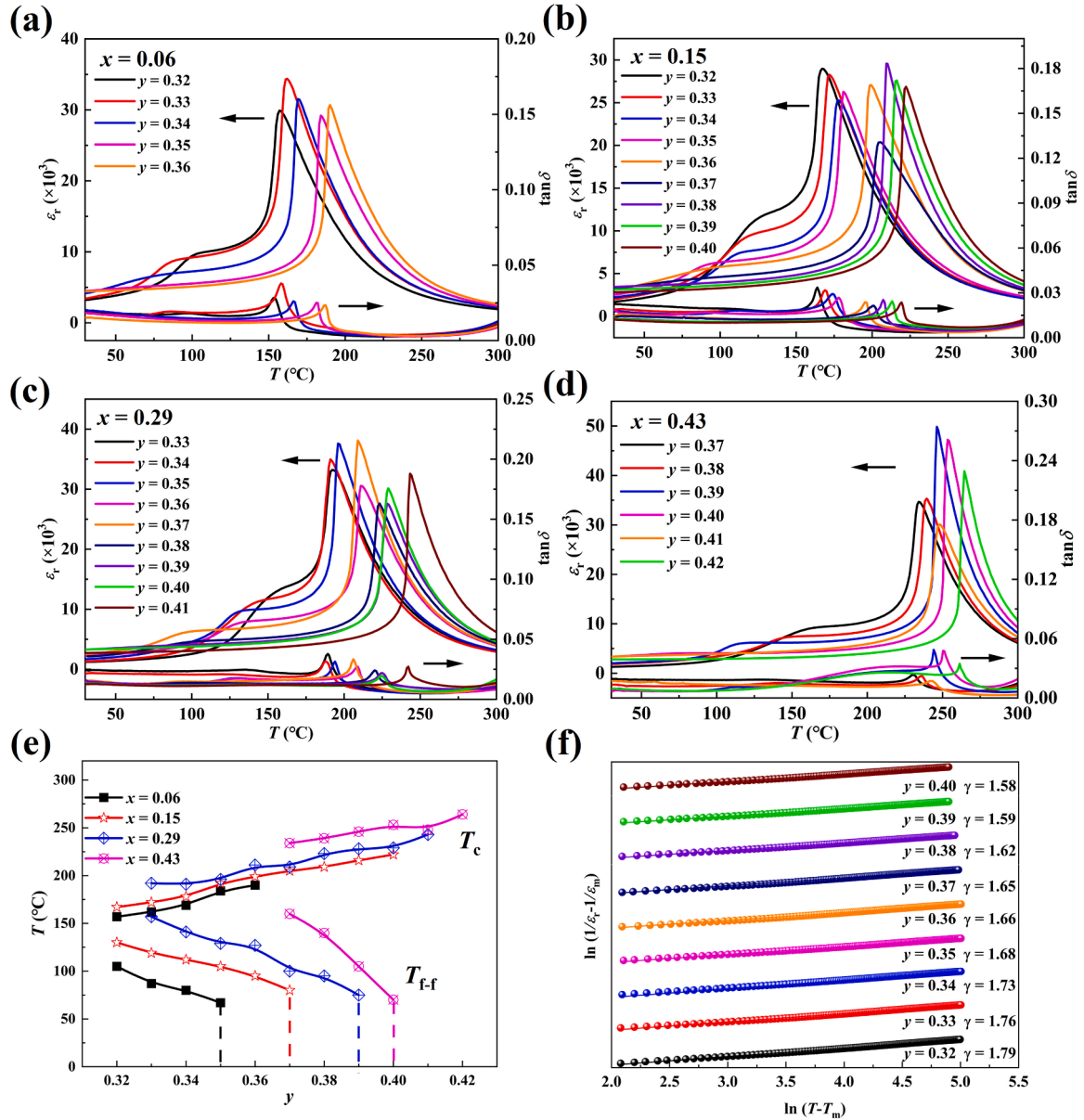


Fig. 6. The dielectric constant and loss of $x\text{PSN}-(1-x-y)\text{PMN}-y\text{PT}$ ceramics as a function of the temperature: (a) $x = 0.06$, (b) $x = 0.15$, (c) $x = 0.29$, (d) $x = 0.43$, (e) The T_c and T_{f-f} of $x\text{PSN}-(1-x-y)\text{PMN}-y\text{PT}$ ceramics, (f) $\ln(1/\epsilon_r - 1/\epsilon_m)$ as a function of $\ln(T - T_m)$ at 1 kHz for $x = 0.15$ samples.

obtained at $y = 0.36, 0.37$ mainly results from the easy polarization rotation induced by external electric field owing to the appropriate phase fraction of $T + M_C$. As the PT content is further increased, d_{33} shows a marked decrease. Combining the refinement results, the decrease of d_{33} can be attributed to the further increasing fraction of the tetragonal phase proportion, because the domain switching gets harder. Moreover, as the x increases, the component with the highest d_{33} shifts to the region with higher PT content, which is due to the larger rhombohedral lattice deformation of PSN. The k_p shows a similar trend to that of d_{33} , which exhibits an initial increase followed by a subsequent decrease as the y increases, and the k_p value can reach up to 0.67, as depicted in Fig. S3. Table 1 lists the detailed d_{33} and k_p values for each ceramic.

The piezoelectric performances of piezoceramics are subject to temperature variation due to the thermal depolarization. Fig. 8 presents the temperature-dependent pyroelectric current density for the poled 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT ceramics. We can

obviously see that each sample exhibits two current density peaks and the corresponding temperatures are 70 $^\circ\text{C}$ and 181 $^\circ\text{C}$ for 0.06PSN-0.59PMN-0.35PT, 75 $^\circ\text{C}$ and 200 $^\circ\text{C}$ for 0.15PSN-0.48PMN-0.37PT, 75 $^\circ\text{C}$ and 226 $^\circ\text{C}$ for 0.29PSN-0.32PMN-0.39PT, 68 $^\circ\text{C}$ and 248 $^\circ\text{C}$ for 0.43PSN-0.17PMN-0.40PT samples. These temperatures are almost in agreement with their T_{f-f} and T_c obtained from $\epsilon_r - T$ curves. The pyroelectric current peak is in general caused by the phase transition. Obviously, the current peak connected with F-F transition for each ceramic is flatter, signifying the good polarization stability. Fig. 9(a) and (b) present the temperature dependence of in-situ $k_{33} T/k_{33} \text{ RT}$ and $d_{33} T/d_{33} \text{ RT}$ for the poled 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT samples. The $k_{33} T/k_{33} \text{ RT}$ values of all ceramics display a trend of monotone decrease as the temperature increases. The $k_{33} T/k_{33} \text{ RT}$ values of all samples reduce by within 10 % in the test temperature range from 30 to 100 $^\circ\text{C}$. When the test temperature increases to around T_c of each ceramic, the $k_{33} T/k_{33} \text{ RT}$ only decreases 29 %, 24 %, 15 %, and 24 % for 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-

Table 1

Dielectric, piezoelectric and ferroelectric properties of x PSN-(1 - x - y)PMN- y PT ceramics.

Composition (PSN-PMN-PT)	d_{33} (pC/N)	k_p	T_c (°C)	T_{ft} (°C)	ϵ_r	$\tan\delta$	E_c (kV/cm)	P_r ($\mu\text{C}/\text{cm}^2$)
0.06-0.62-0.32	560	0.61	157	105	3204	0.019	4.9	30.6
0.06-0.61-0.33	570	0.62	162	87	3413	0.018	5.6	29.5
0.06-0.60-0.34	580	0.61	169	80	4160	0.020	6.9	31.7
0.06-0.59-0.35	665	0.62	184	67	4473	0.019	7.8	28.6
0.06-0.58-0.36	642	0.59	190	-	4423	0.015	7.7	29.1
0.15-0.53-0.32	405	0.56	167	130	1587	0.021	5.2	28.9
0.15-0.52-0.33	500	0.63	172	119	2016	0.019	5.5	32.8
0.15-0.51-0.34	529	0.64	178	112	2227	0.017	6.1	32.5
0.15-0.50-0.35	540	0.64	181	105	2519	0.015	6.4	31.7
0.15-0.49-0.36	575	0.64	199	95	3144	0.014	6.6	33.0
0.15-0.48-0.37	620	0.60	205	80	3309	0.014	7.3	28.6
0.15-0.47-0.38	545	0.57	209	-	3456	0.013	8.7	28.0
0.15-0.46-0.39	478	0.54	216	-	3104	0.013	10.0	26.8
0.15-0.45-0.40	404	0.52	222	-	2792	0.013	11.3	26.9
0.29-0.38-0.33	335	0.55	192	157	1213	0.024	5.5	34.4
0.29-0.37-0.34	365	0.57	191	141	1288	0.022	5.5	34.6
0.29-0.36-0.35	460	0.64	196	129	1678	0.014	6.5	32.6
0.29-0.35-0.36	535	0.67	211	127	2220	0.014	6.5	30.6
0.29-0.34-0.37	540	0.66	209	100	2525	0.015	6.5	34.8
0.29-0.33-0.38	545	0.63	223	95	2610	0.013	7.0	32.7
0.29-0.32-0.39	585	0.61	228	75	3297	0.016	8.8	29.2
0.29-0.31-0.40	575	0.60	229	-	3290	0.013	10.0	27.5
0.29-0.30-0.41	467	0.55	243	-	2718	0.013	11.8	28.9
0.43-0.20-0.37	340	0.58	234	160	1300	0.019	5.5	36.1
0.43-0.19-0.38	490	0.67	239	140	1866	0.013	7.0	35.4
0.43-0.18-0.39	520	0.67	246	105	2032	0.009	7.0	36.0
0.43-0.17-0.40	650	0.64	253	70	3347	0.007	8.5	31.0
0.43-0.16-0.41	570	0.58	248	-	3360	0.017	9.2	32.3
0.43-0.15-0.42	475	0.53	264	-	2774	0.011	11.7	28.8

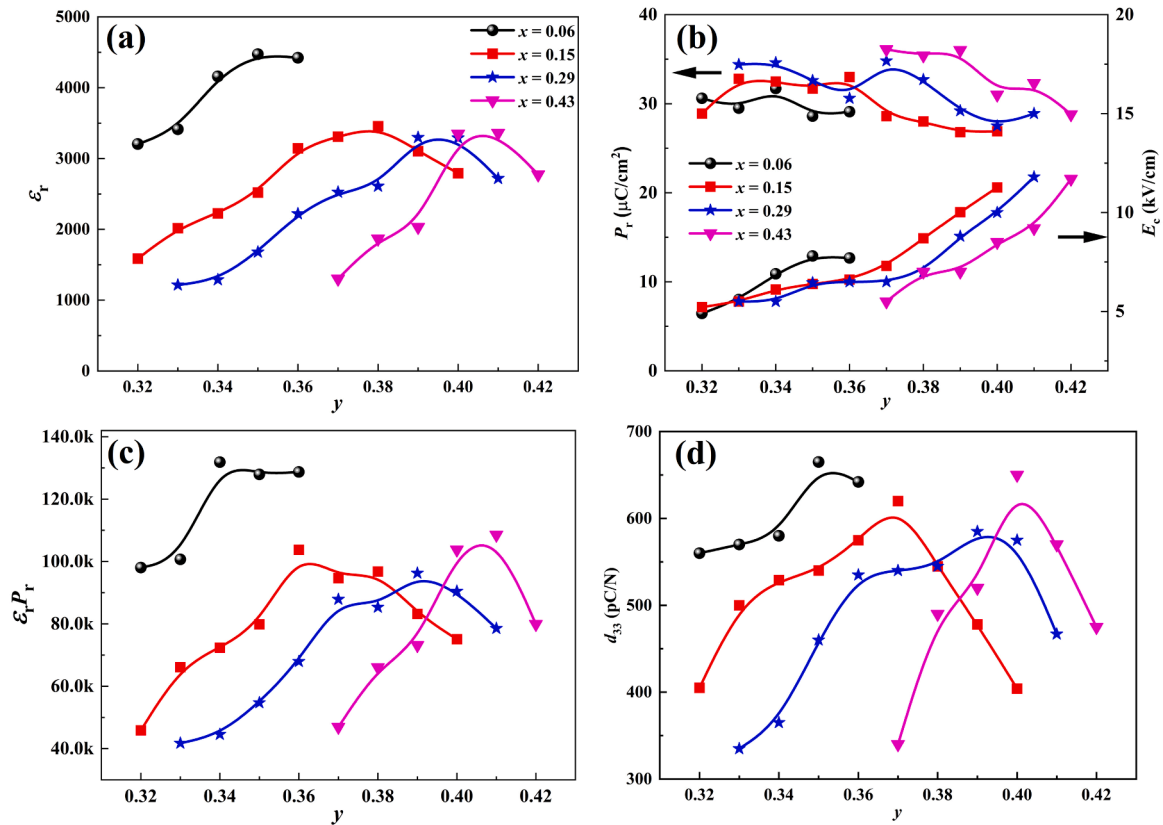


Fig. 7. (a) The dielectric constant, (b) E_c and P_r , (c) $\epsilon_r P_r$ and (d) d_{33} of x PSN-(1 - x - y) PMN- y PT ceramics as a function of PT.

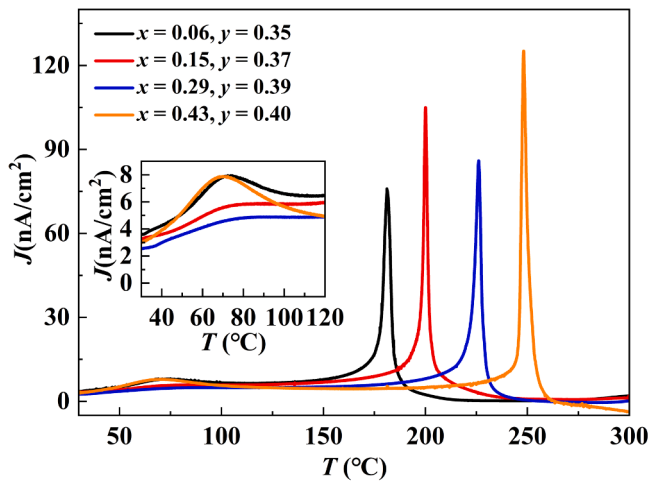


Fig. 8. The temperature-dependent pyroelectric current density of the poled xPSN- (1 - x - y)PMN-yPT ceramics.

0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT ceramics, respectively, demonstrating the k_{33} possesses good temperature stability. The in-suit d_{33} is calculated by the formula [41]: $d_{33} = k_{33} * (\epsilon_{33}^T / s_{33}^E)^{1/2}$, where ϵ_{33}^T is the free dielectric constant, s_{33}^E refers to the elastic compliance coefficient. The temperature dependence of ϵ_{33}^T and s_{33}^E results of xPSN-(1 - x - y)PMN-yPT ceramics are depicted in Fig. S4. As presented in Fig. 9 (b), as the temperature increases, the variation trend of d_{33T}/d_{33RT} for each component is similar, showing an increase first and then a decrease near T_{fT} and T_c . Other piezoceramic materials have shown similar phenomena as well [41,42]. The temperature at which the d_{33} value suddenly drops commonly refers to the depolarization temperature (T_d) [43]. The T_d values of the ceramic samples are 170 °C ($x/y = 0.06/0.35$), 190 °C ($x/y = 0.15/0.37$), 200 °C ($x/y = 0.29/0.39$) and 230 °C ($x/y = 0.43/0.40$). From room temperature to the T_d of each ceramic, the variation rate of d_{33} is 13 %, 7 %, 5 % and 15 % for 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT ceramics, respectively. The results mentioned above mean that the PSN-PMN-PT ceramics exhibit a good thermal stability.

In order to evaluate the potential of PSN-PMN-PT ceramics in ultrasonic transducers, the 0.06PSN-0.59PMN-0.35PT sample with the highest d_{33} of 665 pC/N is chosen as an active element of ultrasonic transducer. Fig. 10(b) and (c) displays the simulated and tested pulse-echo responses and Fourier spectra for the fabricated PSN-PMN-PT

transducer, respectively. It can be seen that the measured f_c and -6 dB bandwidth are 4.65 MHz and 77.4 %, respectively, which agree well with the simulated results ($f_c \sim 4.69$ MHz and $BW \sim 80.3$ %). The -6 dB bandwidth of the PSN-PMN-PT transducer surpasses that of other transducers reported in the literature with the similar center frequency [44,45], as shown in Fig. 10(d). The IL of the transducer measured at f_c is -21.45 dB, which is better than the transducer fabricated using PMN-PT single crystal ($IL \sim -28.7$ dB) [45]. According to the obtained data, it can be concluded that PSN-PMN-PT ceramics exhibit great potential for ultrasonic transducer applications.

4. Conclusions

In summary, xPSN-(1 - x - y)PMN-yPT ($x = 0.06 - 0.43$, $y = 0.32 - 0.42$) ceramics were fabricated with different phase structure. Comprehensive research has been done on the relationships between composition, phase structure, electrical properties and thermal stability. The consequences revealed that as the PT content rises, the ceramics with identical PSN content demonstrate a phase transition sequence of $M_C + R \rightarrow M_C + M_A + R \rightarrow M_C + T$. Meanwhile, as the PSN content increases, the MPB compositions exhibit a shift toward higher PT content. The maximum d_{33} value for each x is obtained at 0.06PSN-0.59PMN-0.35PT, 0.15PSN-0.48PMN-0.37PT, 0.29PSN-0.32PMN-0.39PT and 0.43PSN-0.17PMN-0.40PT ceramics, in which both T and M_C phases coexist but the T phase is dominant, and the corresponding value is 665, 620, 585, and 650 pC/N, respectively. The d_{33} variation rates of these ceramics, ranging from room temperature to the T_d of each ceramic, are 13 %, 7 %, 5 %, and 15 %, respectively, indicating that these ceramics have a good thermal stability. Moreover, the ultrasonic transducer ($f_c \sim 4.69$ MHz) was prepared using the 0.06PSN-0.59PMN-0.35PT ceramic with the highest d_{33} as an active element. The -6 dB bandwidth is 77.4 % and two-way insertion loss is -21.45 dB. The findings of this study have guiding significance for the advancement of high-performance PT-based ferroelectric ceramics.

CRediT authorship contribution statement

Xiaoqing Huo: Writing – original draft, Validation, Formal analysis, Conceptualization. **Zichao Zhao:** Validation, Data curation. **Qiyu Ge:** Validation, Data curation. **Feifei Guo:** Writing – review & editing, Project administration, Formal analysis, Conceptualization. **Hongqiao Zhou:** Writing – original draft, Validation, Formal analysis, Conceptualization. **Wei Long:** Supervision, Project administration, Methodology. **Zhonghua Dai:** Supervision, Methodology. **Zengzhe Xi:** Writing – review & editing, Resources, Project administration, Funding acquisition.

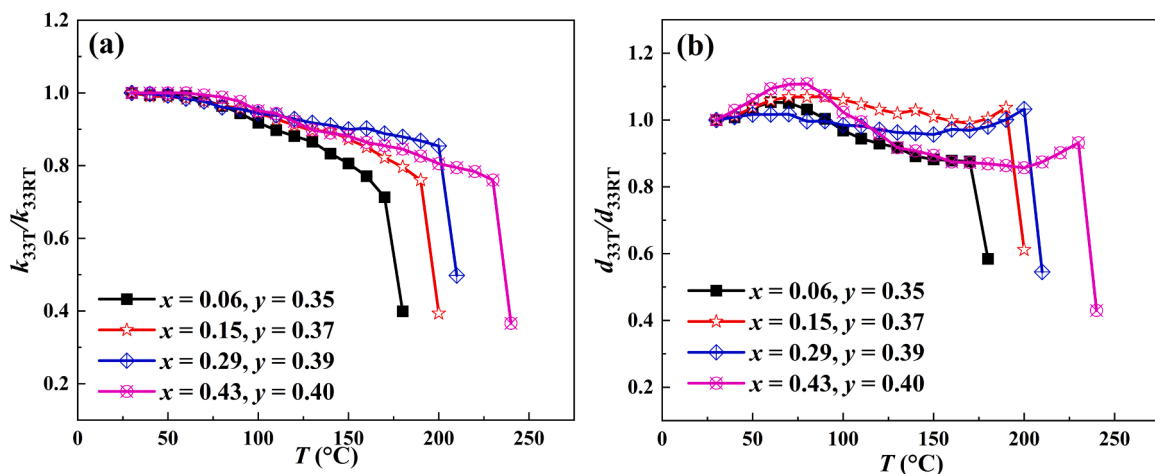


Fig. 9. The temperature dependence of (a) k_{33T}/k_{33RT} and (b) d_{33T}/d_{33RT} values of xPSN- (1 - x - y)PMN-yPT ceramics [20].

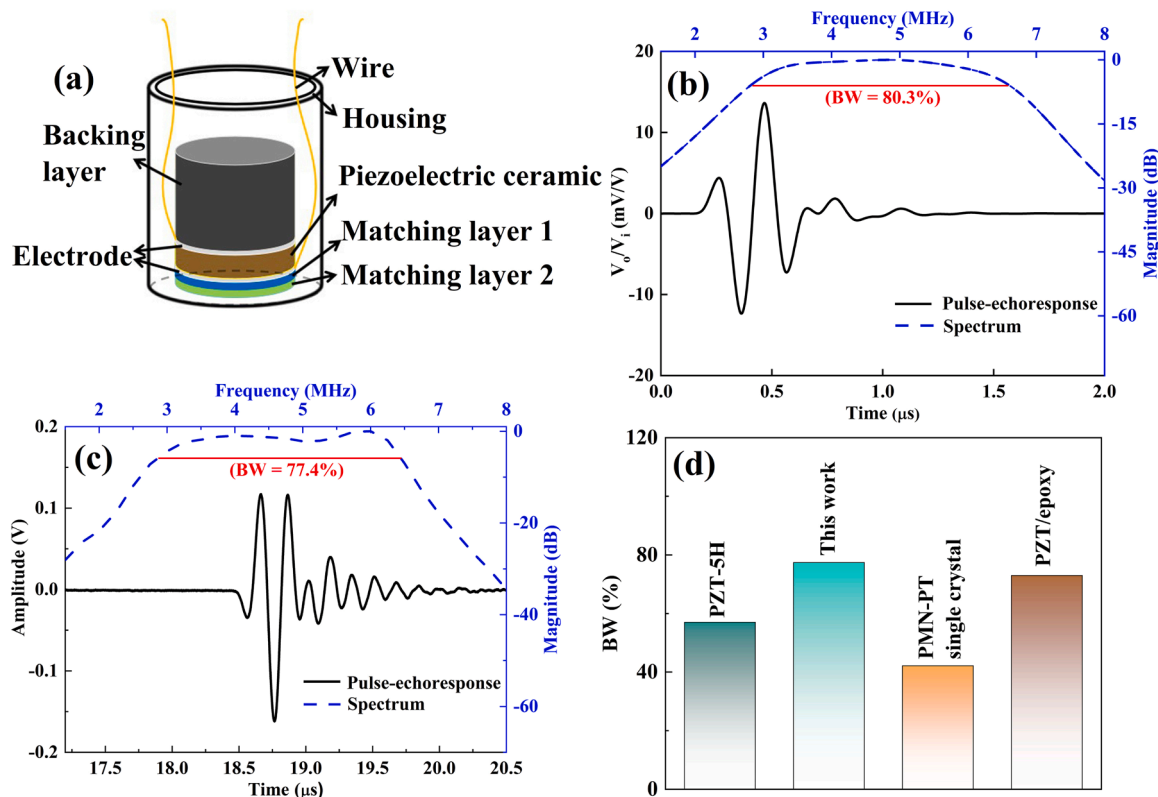


Fig. 10. (a) Structure diagram of ultrasonic transducer, (b) The simulated and (c) measured pulse-echo responses and Fourier spectra for the transducer, (d) Performance of transducers with the similar center frequency.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2025.178686](https://doi.org/10.1016/j.jallcom.2025.178686).

Data availability

Data will be made available on request.

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